



National Institute of
Diabetes and Digestive
and Kidney Diseases

Type 2 Diabetes

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What is type 2 diabetes?

Type 2 diabetes, the most common type of diabetes, is a disease that occurs when your blood glucose, also called blood sugar, is too high. Blood glucose is your main source of energy and comes mainly from the food you eat. [Insulin](#), a [hormone](#) made by the [pancreas](#), helps glucose get into your cells to be used for energy. In type 2 diabetes, your body doesn't make enough insulin or doesn't use insulin well. Too much glucose then stays in your blood, and not enough reaches your cells.

The good news is that you can take steps to prevent or delay the development of type 2 diabetes.

Who is more likely to develop type 2 diabetes?

You can develop type 2 diabetes at any age, even during childhood. However, type 2 diabetes occurs most often in middle-aged and older people. You are more likely to develop type 2 diabetes if you are age 45 or older, have a family history of diabetes, or are [overweight](#) or have [obesity](#). Diabetes is more common in people who are African American, Hispanic/Latino, American Indian, Asian American, or Pacific Islander.

Physical inactivity and certain health problems such as [high blood pressure](#) affect your chances of developing type 2 diabetes. You are also more likely to develop type 2 diabetes if you have [prediabetes](#) or had [gestational diabetes](#) when you were pregnant. Learn more about [risk factors for type 2 diabetes](#).



Type 2 diabetes occurs most often in middle-aged and older people.

What are the symptoms of diabetes?

Symptoms of diabetes include

- increased thirst and urination
- increased hunger
- feeling tired
- blurred vision
- numbness or tingling in the feet or hands
- sores that do not heal
- unexplained weight loss

Symptoms of type 2 diabetes often develop slowly—over the course of several years—and can be so mild that you might not even notice them. Many people have no symptoms. Some people do not find out they have the disease until they have diabetes-related health problems, such as blurred vision or [heart disease](#).

What causes type 2 diabetes?

Type 2 diabetes is caused by several factors, including

- overweight and obesity
- not being physically active
- [insulin resistance](#)
- [genes](#)

Learn more about the [causes of type 2 diabetes](#).

How do health care professionals diagnose type 2 diabetes?

Your health care professional can diagnose type 2 diabetes based on blood tests. Learn more about [blood tests for diabetes](#) and what the results mean.

How can I manage my type 2 diabetes?

Managing your blood glucose, [blood pressure](#), and [cholesterol](#), and quitting smoking if you smoke, are important ways to **manage your type 2 diabetes**. Lifestyle changes that include planning healthy meals, limiting calories if you are overweight, and being physically active are also part of managing your diabetes. So is taking any prescribed medicines. Work with your health care team to create a diabetes care plan that works for you.



Following your meal plan helps you manage your diabetes.

What medicines do I need to treat my type 2 diabetes?

Along with following your diabetes care plan, you may need diabetes medicines, which may include pills or medicines you inject under your skin, such as insulin. Over time, you may need more than one diabetes medicine to manage your blood glucose. Even if you don't take insulin, you may need it at special times, such as during pregnancy or if you are in the hospital. You also may need medicines for high blood pressure, high cholesterol, or other conditions.

Learn more about [medicines, insulin, and other diabetes treatments](#).

What health problems can people with diabetes develop?

Following a good diabetes care plan can help protect against many diabetes-related health problems. However, if not managed, diabetes can lead to problems such as

- heart disease and [stroke](#)
- nerve damage
- kidney disease
- foot problems
- eye disease
- gum disease and other dental problems
- sexual and bladder problems

Many people with type 2 diabetes also have [nonalcoholic fatty liver disease \(NAFLD\)](#). Losing weight if you are overweight or have obesity can improve NAFLD. Diabetes is also linked to other health problems such as [sleep apnea](#), depression, some types of cancer, and [dementia](#) [NIH](#) [↗](#).

You can take steps to lower your chances of developing these [diabetes-related health problems](#).

How can I lower my chances of developing type 2 diabetes?

Research such as the [Diabetes Prevention Program](#) [↗](#), sponsored by the National Institutes of Health, has shown that you can take steps to reduce your chances of developing type 2 diabetes if you have risk factors for the disease. Here are some things you can do to lower your risk:

- **Lose weight if you are overweight, and keep it off.** You may be able to prevent or delay diabetes by losing 5 to 7 percent of your current weight.¹ For instance, if you weigh 200 pounds, your goal would be to lose about 10 to 14 pounds.

Diabetes Risk Management Calculator

Losing 5% to 7% of your body weight may reduce your risk of diabetes.[†]

Height

Weight

Calculate

† For adults 20 and older at high risk of developing type 2 diabetes

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- **Move more.** Get at least 30 minutes of physical activity, such as walking, at least 5 days a week. If you have not been active, talk with your health care professional about which activities are best. Start slowly and build up to your goal.
- **Eat healthy foods.** Eat smaller portions to reduce the amount of calories you eat each day and help you lose weight. Choosing foods with less fat is another way to reduce calories. Drink water instead of sweetened beverages.

Ask your health care team what other changes you can make to prevent or delay type 2 diabetes.

Most often, your best chance for preventing type 2 diabetes is to make lifestyle changes that work for you long term. Get started with [Your Game Plan to Prevent Type 2 Diabetes](#).

References

[1] Diabetes Prevention Program Research Group. Long-term effects of lifestyle intervention or metformin on diabetes development and microvascular complications over 15-year follow-up: the Diabetes Prevention Program Outcomes Study. *The Lancet Diabetes & Endocrinology*. 2015;3(11):866–875. You can find more information about this study on the [Diabetes Prevention Program Outcomes Study website](#).

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